25901868 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

second SEMESTER

Skill Development Components (SDC) - MG2SDCREM100 - ENERGY AND POWER FUNDAMENTALS

(2025 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **25** questions

Each question carries **1** marks

1. Identify the purpose of power generation in a power system. [K] / [CO1]
2. State the purpose of power transmission. [K] / [CO1]
3. Classify the type of value measured by an AC voltmeter (Peak or RMS). [K] / [CO1]
4. State Ohm’s Law. [K] / [CO1]
5. State the unit of active power. [K] / [CO1]
6. Describe RMS value of AC voltage. [K] / [CO1]
7. Identify the components required to calculate utilization factor. [U] / [CO2]
8. Identify examples of inductive loads. [U] / [CO2]
9. State the unit of load factor. [U] / [CO2]

10. Identify the components required to calculate demand factor. [U] / [CO2]

11. State the range of power factor. [U] / [CO2]

12. Interpret why reactive power is associated with capacitive loads. [U] / [CO2]

13. Differentiate between the first block and second block in block rate tariff. [An] / [CO3]

14. Interpret why reactive power is absent in a resistive load. [U] / [CO2]

15. Examine the billing pattern in two-part tariff. [An] / [CO3]

16. Compare smart billing and conventional billing systems. [An] / [CO3]

17. Differentiate between analog and digital energy meters. [An] / [CO3]

18. Differentiate between cost-based tariff and incentive-based tariff. [An] / [CO3]

19. Compare analog energy meter and digital energy meter. [An] / [CO3]

20. Why are government policies necessary for improving energy efficiency? [E] / [CO4]

21. Why is star rating important while purchasing appliances? [E] / [CO4]

22. Justify the need for energy efficiency standards in India. [E] / [CO4]

23. Why should industries adopt renewable energy systems? [E] / [CO4]

24. Why is benchmarking important in energy audit? [E] / [CO4]

25. Why is load management necessary for reducing peak demand? [E] / [CO4]

26. Why is walk-through audit suitable for small-scale industries? [E] / [CO4]

27. Why is conservation better than increasing power generation? [E] / [CO4]

[1x25 = 25]

Part B

Short Essay

Answer any **5** questions

Each question carries **5** marks

28. State the importance of distribution system in supplying power to consumers. [K] / [CO1]

29. Explain different forms of power in electrical systems. [K] / [CO1]

30. Explain the importance of improving power factor in electrical systems. [U] / [CO2]

31. Compare monthly load curve and annual load curve [U] / [CO2]

32. Analyze the working principle of Time of Day tariff with a suitable example. [An] / [CO3]

33. Compare maximum demand tariff and two-part tariff. [An] / [CO3]

34. Evaluate the importance of improving power factor in industrial systems. [E] / [CO4]

[5x5 = 25]

Part C

Essay

Answer any **2** questions

Each question carries **10** marks

35. Explain the charging and discharging process of a capacitor in detail. [K] / [CO1]

36. Explain in detail the daily load curve and its significance in power system operation. [U] / [CO2]

37. Examine how a good tariff encourages efficient energy consumption. [An] / [CO3]

38. Assess the importance of each stage in the energy audit procedure. [E] / [CO4]

[10x2 = 20]